ANALYTICAL CUMULATIVE EXAM

February 27, 1999

(4 questions weighted 25 % each)
A popular technique for analyzing clinical samples of dopamine and other compounds found in blood is liquid chromatography with electrochemical detection (LCEC). The most popular electrochemical detectors to date have been based on the amperometric conversion of analyte in a thin-layer cell. The amperometric current produced is proportional to the instantaneous concentration of the analyte in the cell.

In routine applications the electrode is poised at a constant potential so that the resulting current is at the diffusion limit. The general equation for the diffusion limited current is:

$$i = \frac{nFADC}{\delta}$$

The thickness of the diffusion layer δ, is determined by the hydrodynamics of the flowing system. For the case of the thin-layer cell shown below, the value of δ is:

$$\delta = KU^{-1/3}$$

where K mass transport constant that is determined by the properties of the fluid and U is the flow rate in mL/min.

1. A flow system with a thin-layer amperometric detector is used for detection of dopamine. A limiting current peak current of 5.4 μA was recorded for a injection of a 0.1 mM sample of dopamine standard at a flow rate of 1.1 mL/min.

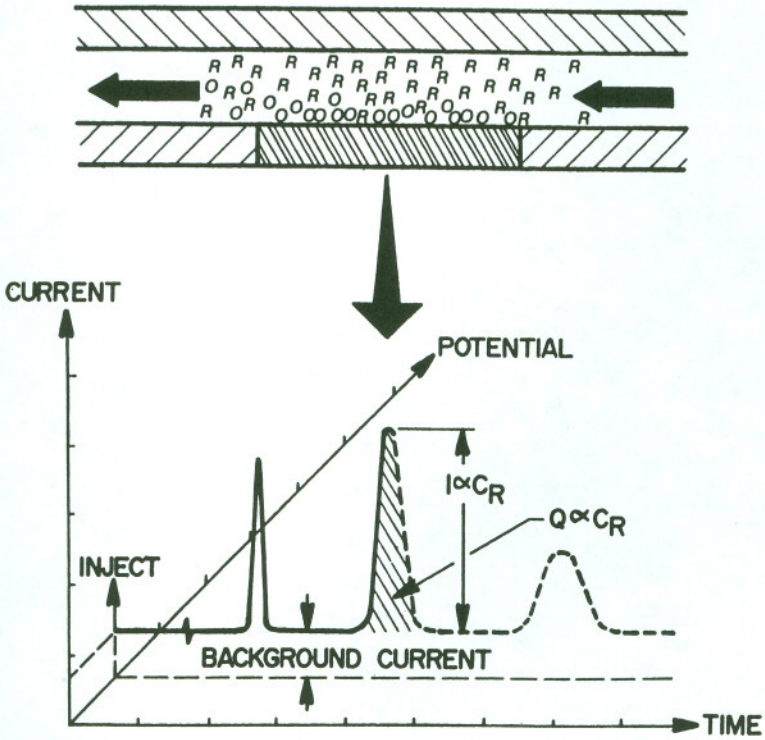
a) The same solution is injected at a flow rate of 2.8 mL/min, By what percentage will the peak current increase?

b) What peak current should be observed for the injection of 0.05 mM dopamine at a flow rate of 1.8 mL/min?

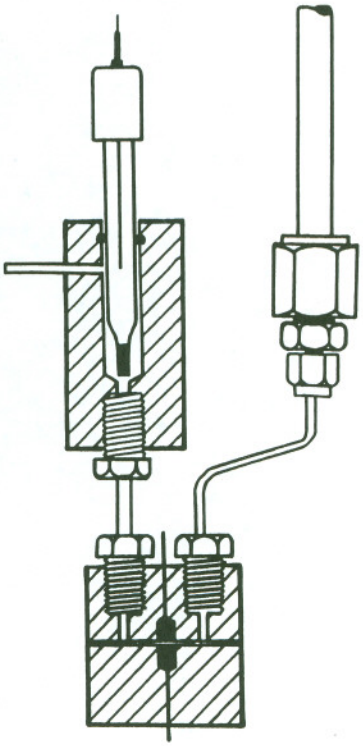
2. The amount of material undergoing electrochemical reaction is only about 0.1% to 5% of the total analyte in the cell. In other words, the vast majority of the sample is not being detected (note the other figure). A "great idea" to improve detection limits would be to get more of the sample to react while present in the cell. This can be accomplished by increasing the size of the electrodes and the current through the cell. Most everybody has given up on this "great idea", and small electrodes converting only a fraction of the analyte in the cell are used almost exclusively. Write a paragraph explaining why this "great idea" failed to improve detection limits.

3. Potential-scanning detectors can increase the information content over the simple fixed potential operation by performing several cyclic voltametric scans as the analyte eludes past the electrode. The resulting data is a three-dimensional detector response: current, against potential and time. Discuss the advantages and disadvantages of the potential-scanning approach compared to the single potential (amperometric) experiment. Make sure you compare the selectivity and sensitivity of each method.

4. LCEC is never do in the "normal phase", Why is this so?



At any instant in time, the current in an LCEC experiment reflects the rate of conversion of reactant (R) to product (O).



Thin-Layer Cell